

Computer Hacking

Forensic Investigator (CHFI)

Version 11

Lab Report

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Course Name: Computer Hacking Forensic Investigator (CHFI V11)

Report Title: CHFI Lab Report (Module 13)

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Table of Contents

Module 13: Email and Social Media Forensics

Lab 1: Investigate a Suspicious Email

Lab 2: Recover Deleted Email Messages Using Recover My Email

Lab 3: Recover Deleted Email Messages Using Recovery Toolbox for Outlook

Lab 4: Extract Information from EML Files Using Python

Module 13: Email and Social Media Forensics

Lab Scenario

Emily had received an email stating that she had won a huge amount from a big company but the amount could only be collected only after paying certain taxes. The email also included instructions along with the account number to send the tax amount to as a precursor to receiving the cash prize. She followed all the instructions in the email and later came to know that it was a spoofed email and someone had scammed her.

To investigate email crimes as a forensic investigator, you must know how to examine the email headers within an email message, trace the origin of a malicious email message, and extract or recover deleted email messages using various email tracking and investigation tools.

Lab Objectives

The objective of this lab is to help you understand how to examine the different components of an email message and other responsibilities that include: Extracting metadata from email headers Finding the origin of a spoofed email message Recovering deleted email messages

Overview of Investigating Email Crimes

Email crime investigation is primarily conducted to examine the content as well as the origin of any email message that is found to be offending or suspected to be spoofed. Investigators need to use different forensic tools to examine emails related to spamming, mail bombing/mail storms, spoofing, phishing attacks, and email hijacking.

Lab Tasks

Recommended labs to assist you in investigating email crimes:

- Investigate a Suspicious Email
- Recover Deleted Email Messages Using Recover My Email
- Recover Deleted Email Messages Using Recovery Toolbox for Outlook
- Extract Information from EML Files Using Python

Lab 1: Investigate a Suspicious Email

Lab Scenario

John Dove, a manager at a reputed hotel, received an email in the hotel's official email account stating that the hotel's Facebook account handle had been compromised and that the password to the account needed to be reset in order to recover the account. The email contained a link and an attachment in the form of a document that supposedly contained the instructions to reset the password for the affected Facebook account. Panicked, John clicked on the link so he could follow the instructions to reset the password for the hotel's Facebook account and recover it. As he went through the process, John realized that something was fishy about the Facebook page he was interacting with on the screen. However, during this process, he also inadvertently divulged the hotel's confidential information. The hotel's management soon learned that its key confidential information had been stolen and was being misused for malicious purposes. John reported his encounter with the fishy email he received that claimed to be from Facebook. Upon learning of the incident, the hotel's management sought the services of a cyber-forensics agency. Investigator Johnson now has to analyze the suspicious email and find its origin.

Lab Objectives

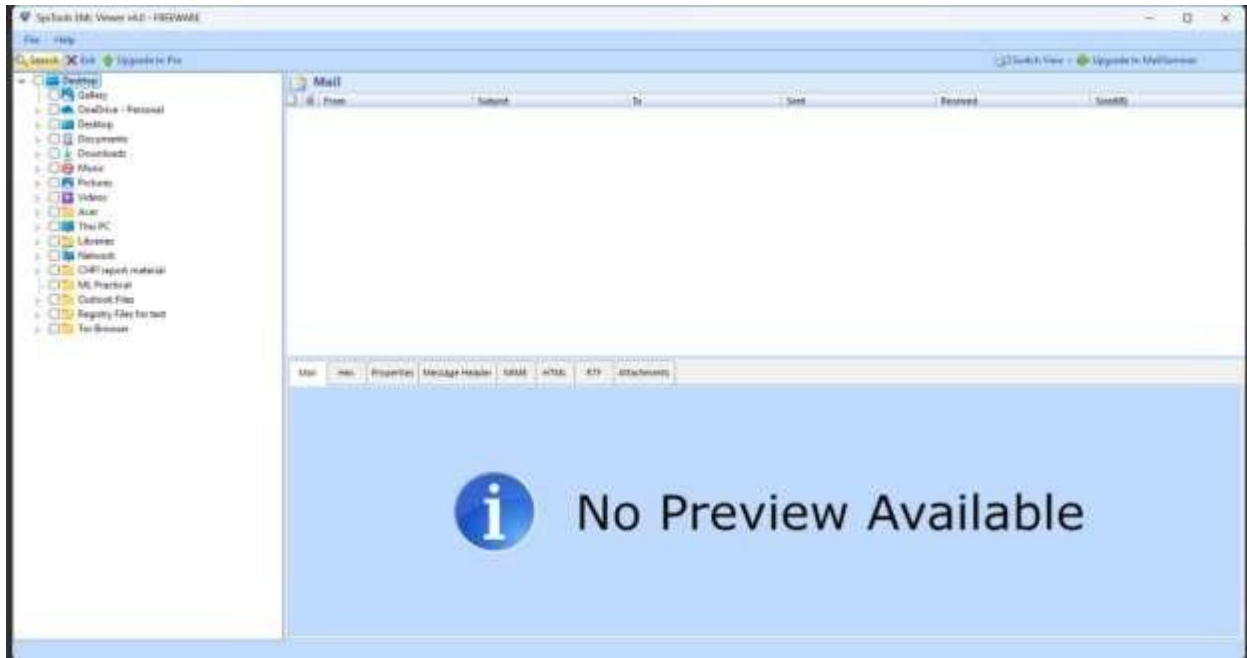
Investigating a suspicious e-mail message involves analyzing its email headers to determine if it is spoofed. It also involves determining the origin of the email message and checking the validity and genuineness of the sender's email address. The objective of this lab is to help you understand how to perform forensic investigation on suspicious emails and analyze them.

Overview of the Lab

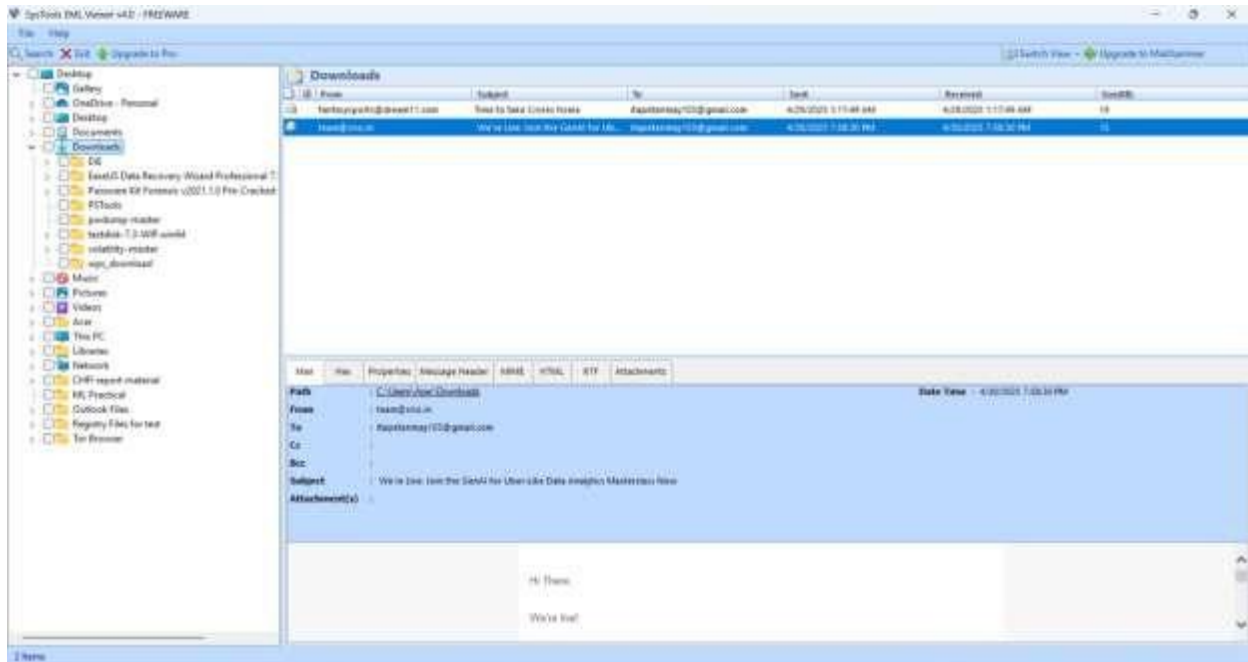
This lab familiarizes you with the process of identifying a suspicious email and examining it in detail to determine if it is spoofed.

Lab Tasks

1. Open **SysTools EML Viewer Pro** Tool as shown in screenshot below:



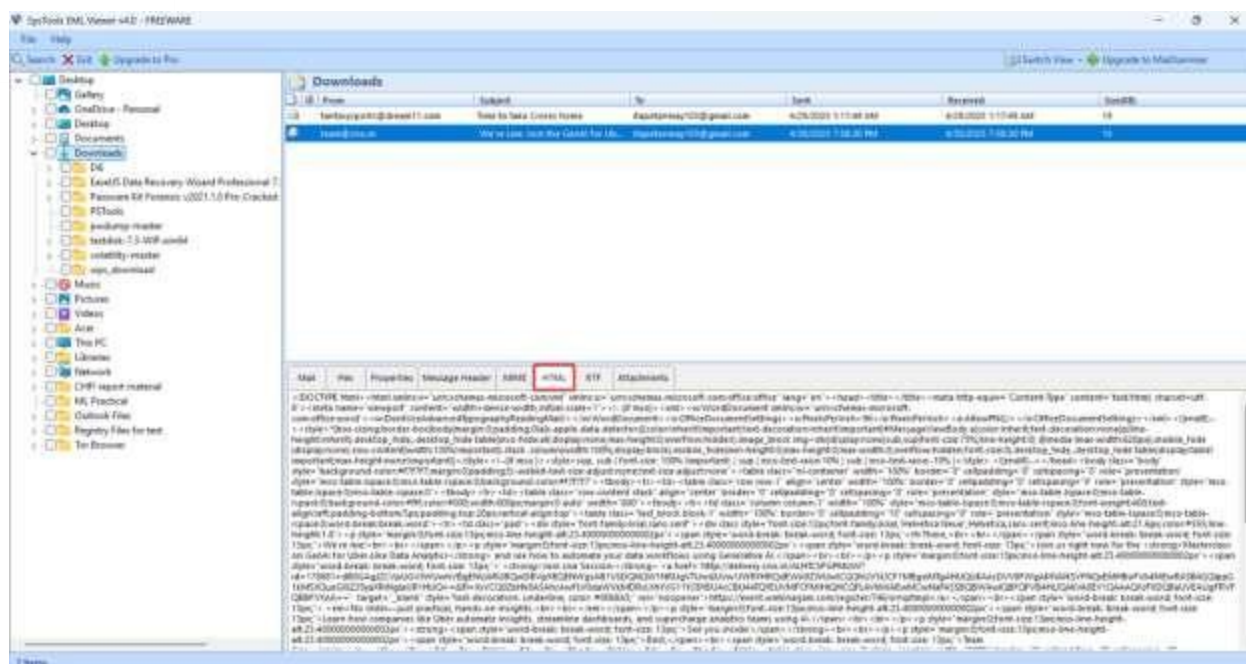
2. Next, in the main window of the application, we need to add our evidence file(s). To add the evidence file(s), in the left pane of the application window, navigate to **Desktop** → **Downloads** by expanding the corresponding nodes and then select the EML Files folder under Evidence Files. Upon selecting the EML Files folder, the .eml files stored within it will be listed in the right pane of the window as indicated in the screenshot below:



3. The contents in the body of an email message can be viewed under the Normal Mail View tab in the lower pane of the application window. Select the second .eml file (Suspicious Email Message file) to view its body contents under the Normal Mail View tab, as indicated in the screenshot below (drag the margin of the lower pane of the window up using the mouse cursor for a full view of the email's contents):

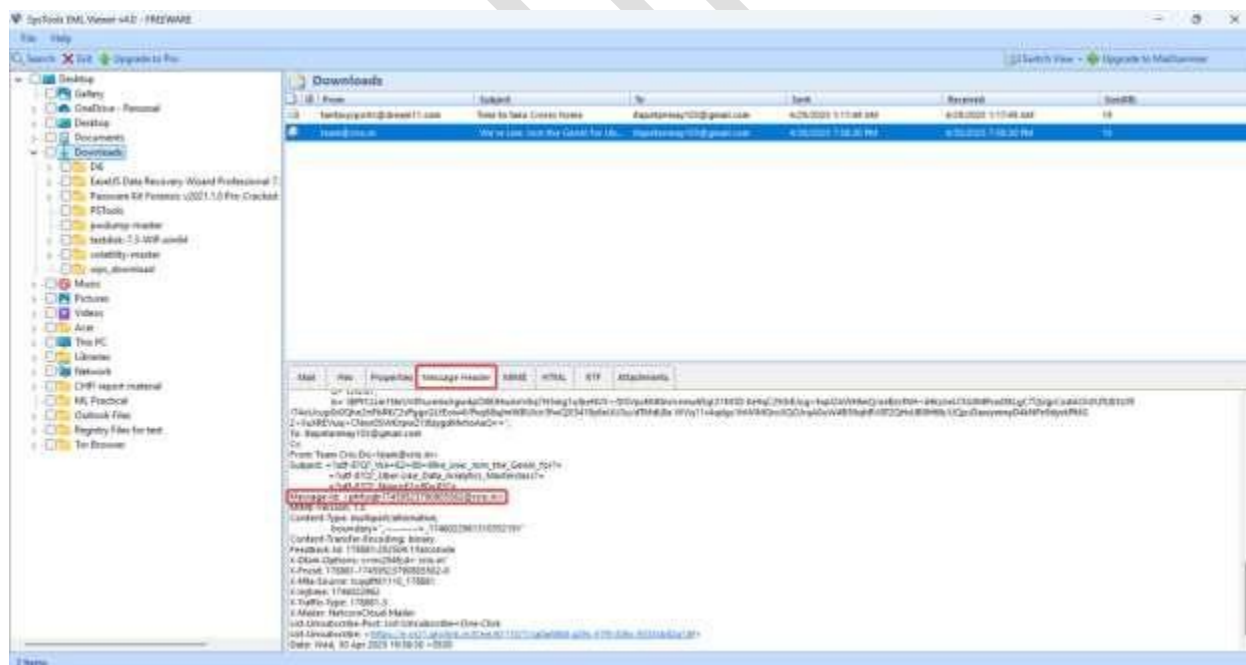
4. Now we shall examine the body of the suspicious email message in an HTML format to establish whether the email message is genuine or malicious.

5. Therefore, ensure that the second .eml file (Suspicious Email Message file) is selected and then click the HTML tab in the lower pane of the application window. The contents of the email message body will now be displayed in an HTML format, as indicated in the screenshot below:

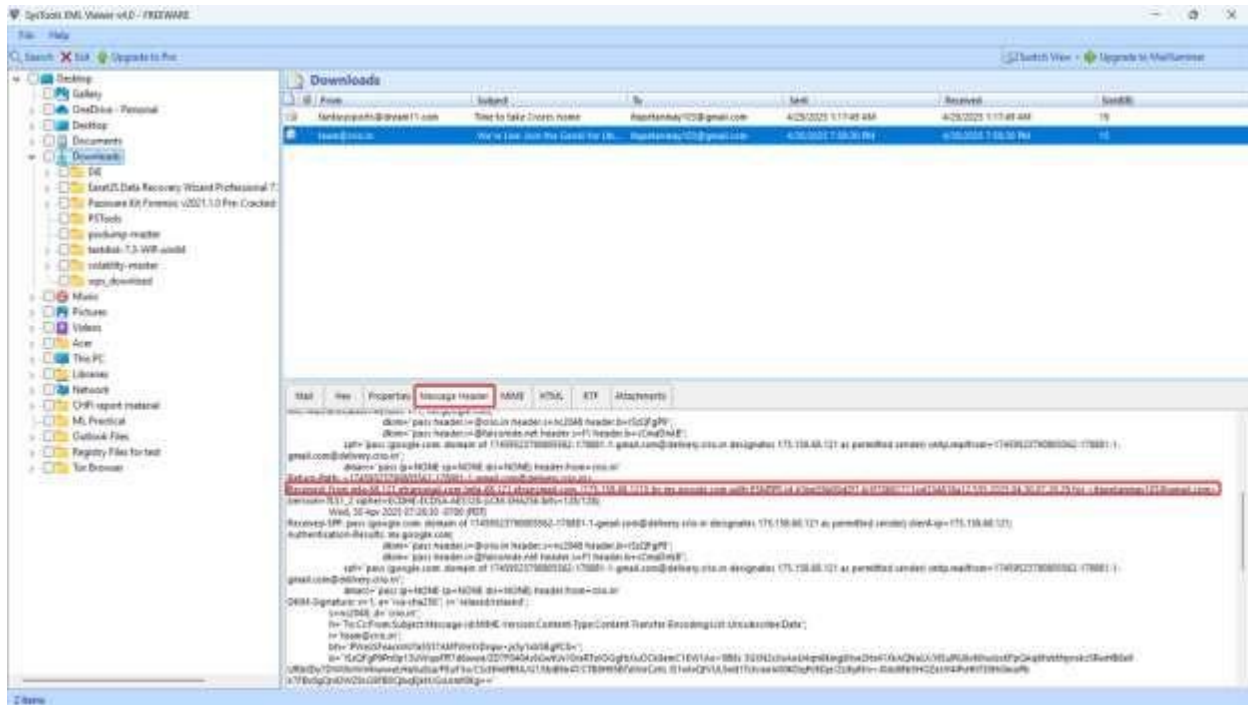


6. Now, we will examine the email headers. Examining the email header information helps establish whether an email message is genuine or spoofed. When using SysTools EML Viewer, all the email header information pertaining to an email message is under the Message Header tab.

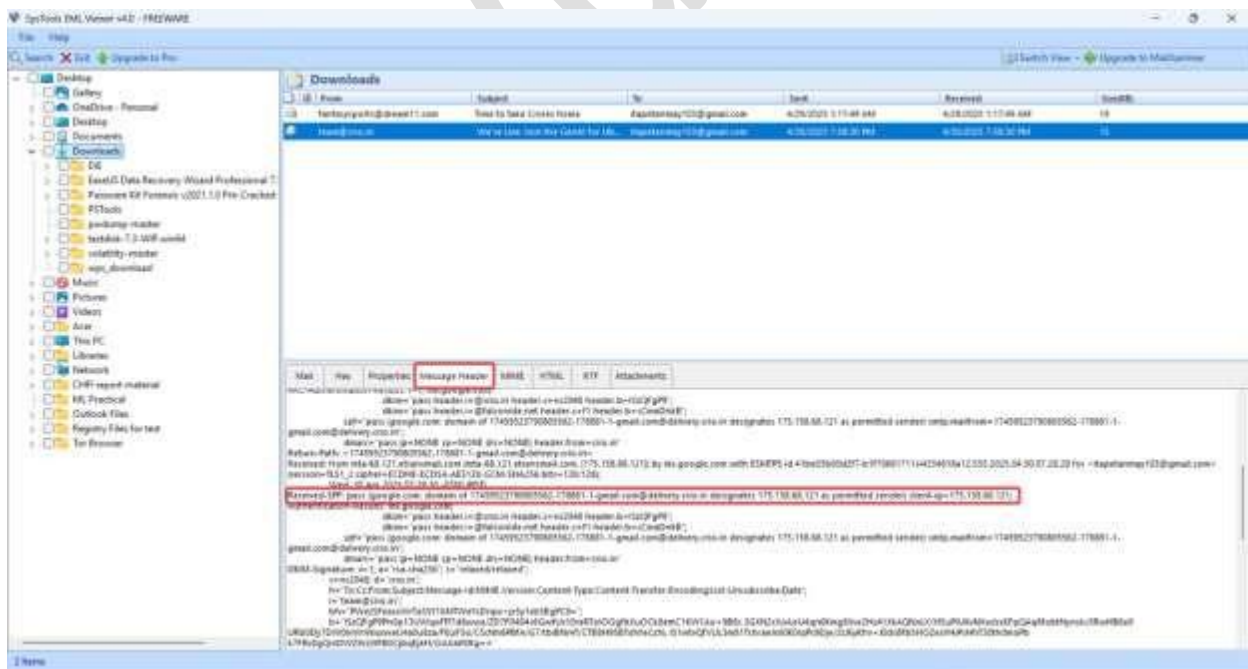
7. We will begin our investigation of email headers by examining the Message ID.



8. Now we will scroll down and see the **Received Header** field displays the domain name **mta-68.121.etransmail.com** and the associated **IP address** from which the email message originated, as shown in the screenshot below:

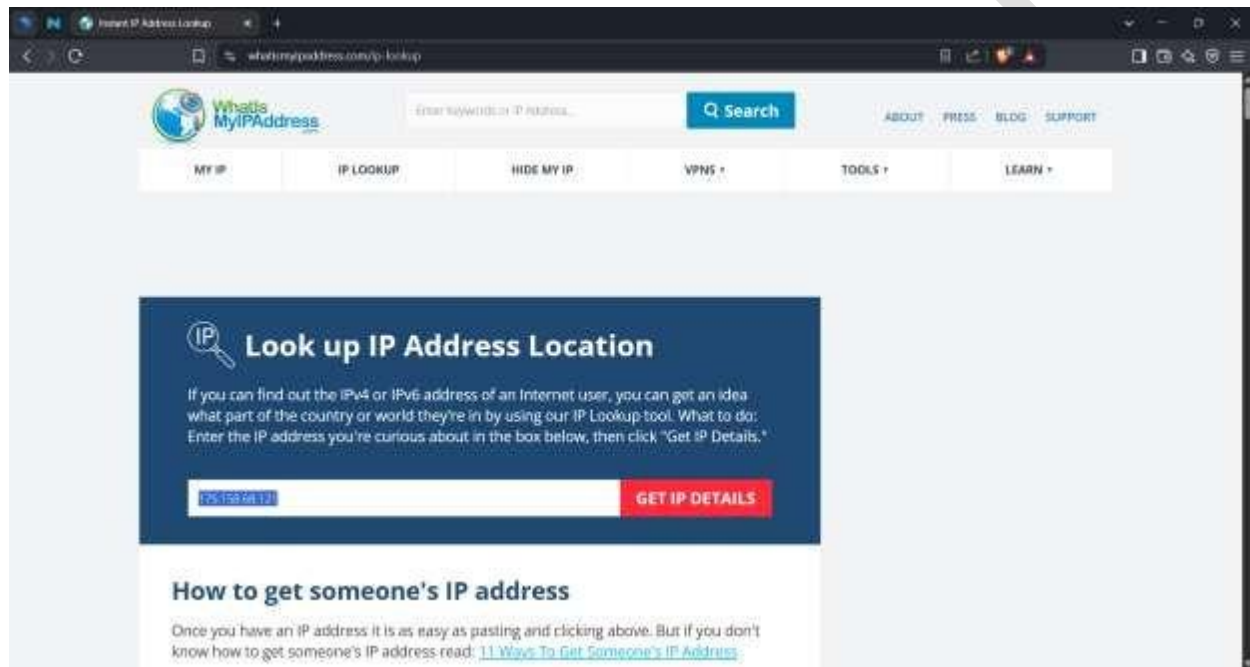


9. We shall now examine the Received-SPF field email messages.

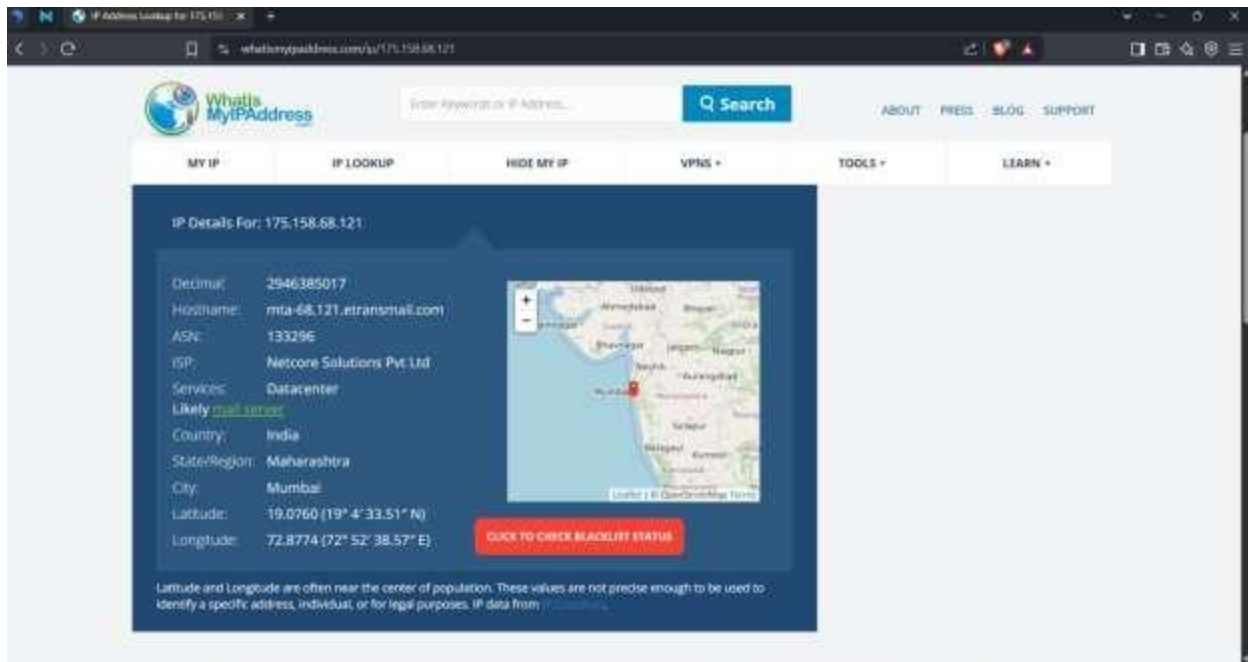


10. Close the tool. Now that we have investigated the critical elements within the spoofed email message, we shall probe its originating IP address on whatismyipaddress.com to determine its geographic origin and the entity that sent it.

11. Launch a browser, type **<https://whatismyipaddress.com/ip-lookup>** in the address bar and press Enter. You will land on the whatismyipaddress.com/ip-lookup webpage where you will find the Get IP Details field. Type the IP address **175.158.68.121** in the field and then click on the Get IP Details button, as indicated in the screenshot below:



12. You will now be redirected to the next page where you will see details pertaining to the IP address 175.158.68.121. As seen in the screenshot below, the IP address is registered with the hostname **mta-68.121.etransmail.com** and it is geographically located in Mumbai.



The hostname might vary in your lab environment.

13. Thus, in this manner, you can investigate a suspicious email, determine if it is spoofed, and find its origin.

14. Close all the windows.

Lab 2: Recover Deleted Email Messages Using Recover My Email

Lab Scenario

Daniel is a web expert and part of a group that hacks other's accounts for monetary benefits. Daniel was a suspect in a case wherein the attacker has used email attachments to send malware to the victim that could copy the user's passwords and other private details and prompt them to the sender. However, when Robert, the forensic investigator dealing with this case, checked Daniel's system, he did not find any incriminating emails. Robert suspected that Daniel might have already deleted those emails. Robert now has to use the correct forensic tool to retrieve the deleted email messages and examine them, if any.

Lab Objectives

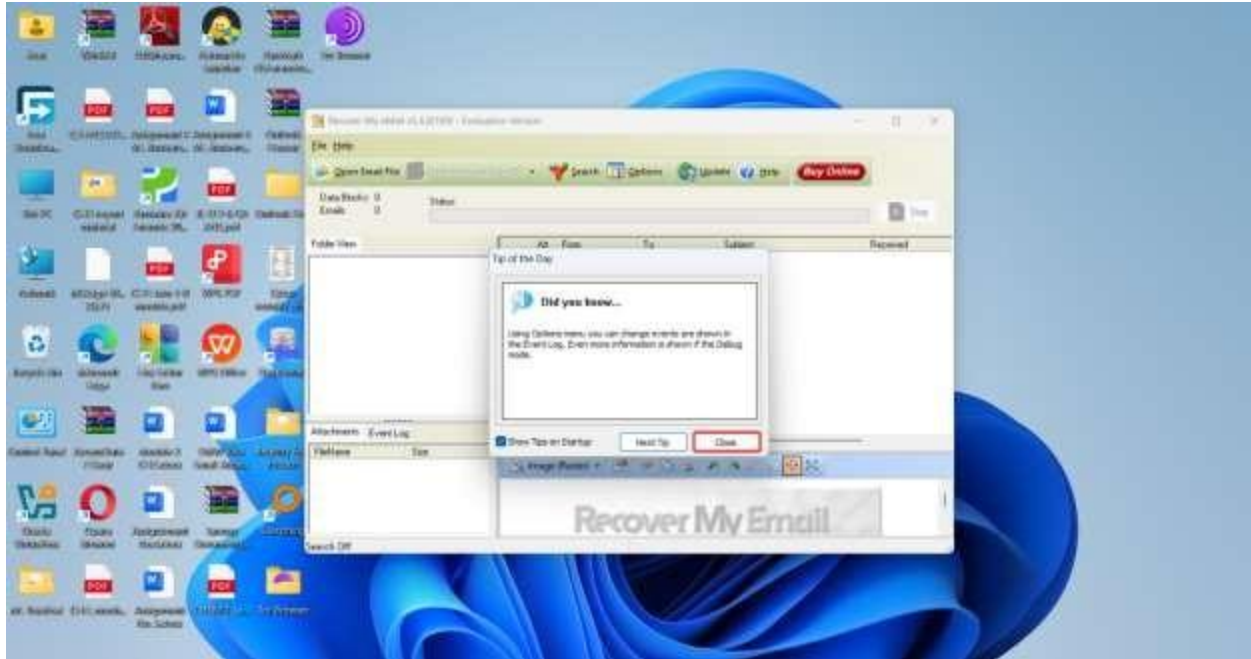
Cybercriminals often delete email messages after committing a crime to avoid being tracked. Investigators can use tools such as Recover My Email, which can retrieve deleted email messages from either Microsoft Outlook PST files or Microsoft Outlook Express DBX files. The objective of this lab is to help investigators understand how to recover deleted email messages and attachments.

Overview of the Lab

This lab familiarizes you with the process of recovering deleted or lost emails with the help of the Recover My Email tool.

Lab Tasks

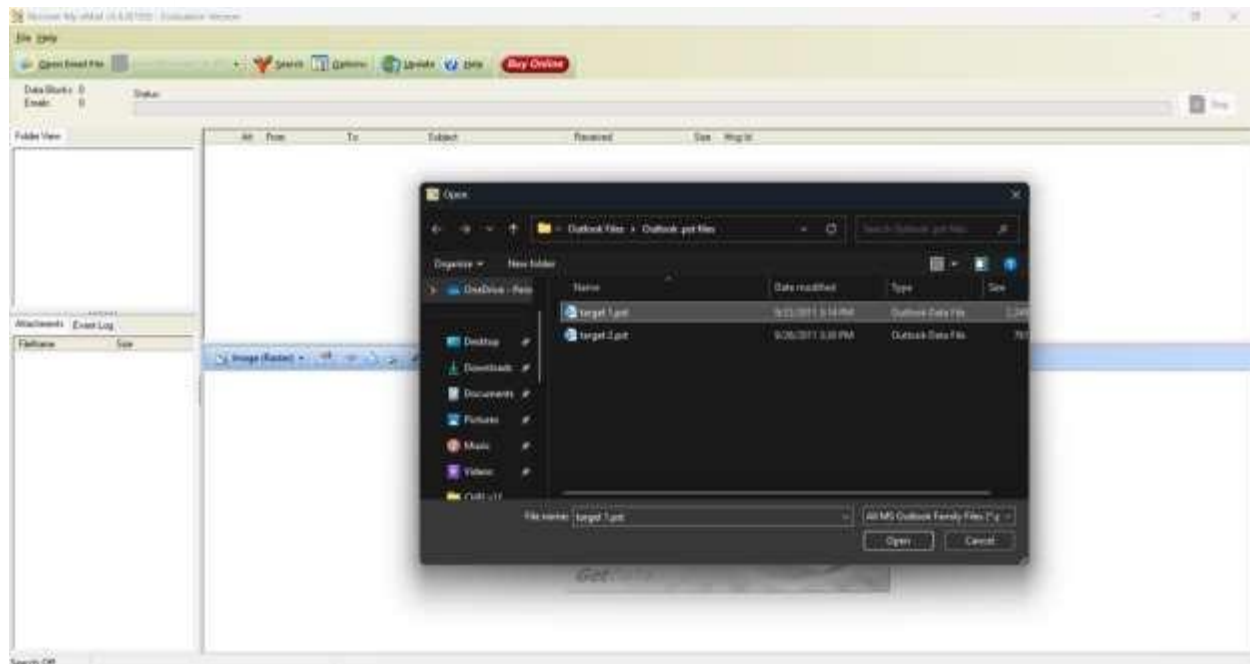
1. In this lab, we will examine the Microsoft Outlook database files for deleted emails.
2. Launch the RecoverMyEmail software and click on **close** if tip-of-the-day window appears.



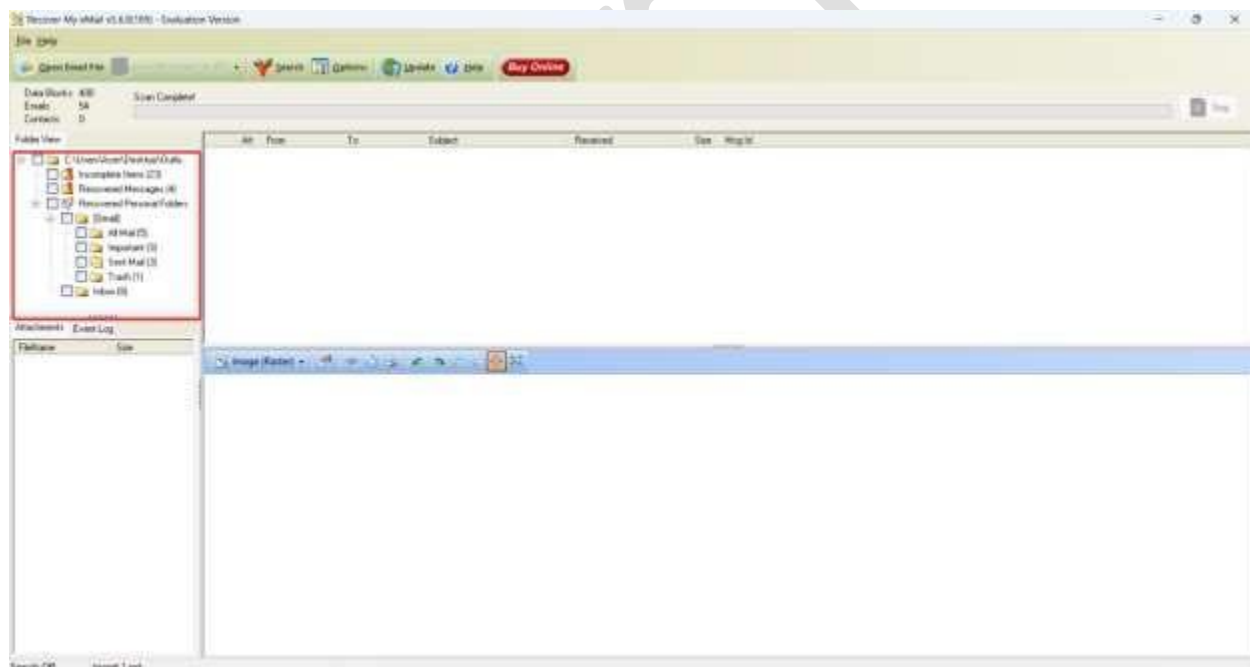
3. Now click on Open Email File as shown below:



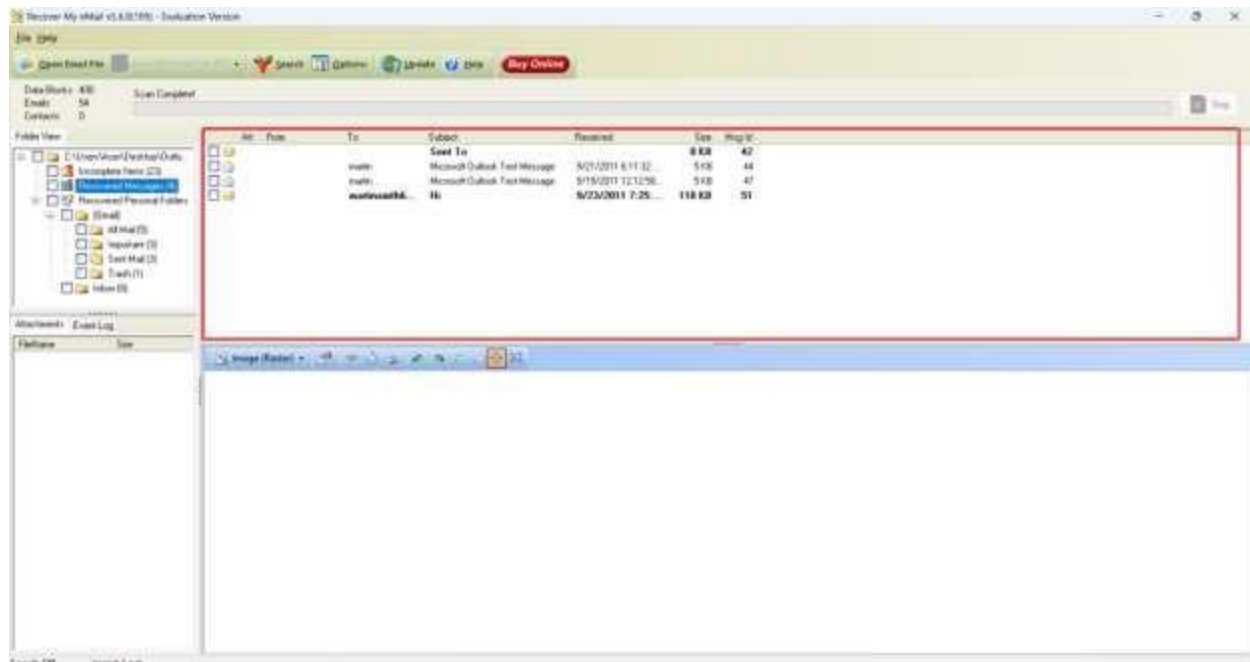
4. Now navigate to Outlook Files → Target pst files and select Target 1.pst and click open.



5. The Recover My Email application will now scan the selected .pst email file and fetch and display the results in the left pane of the window.

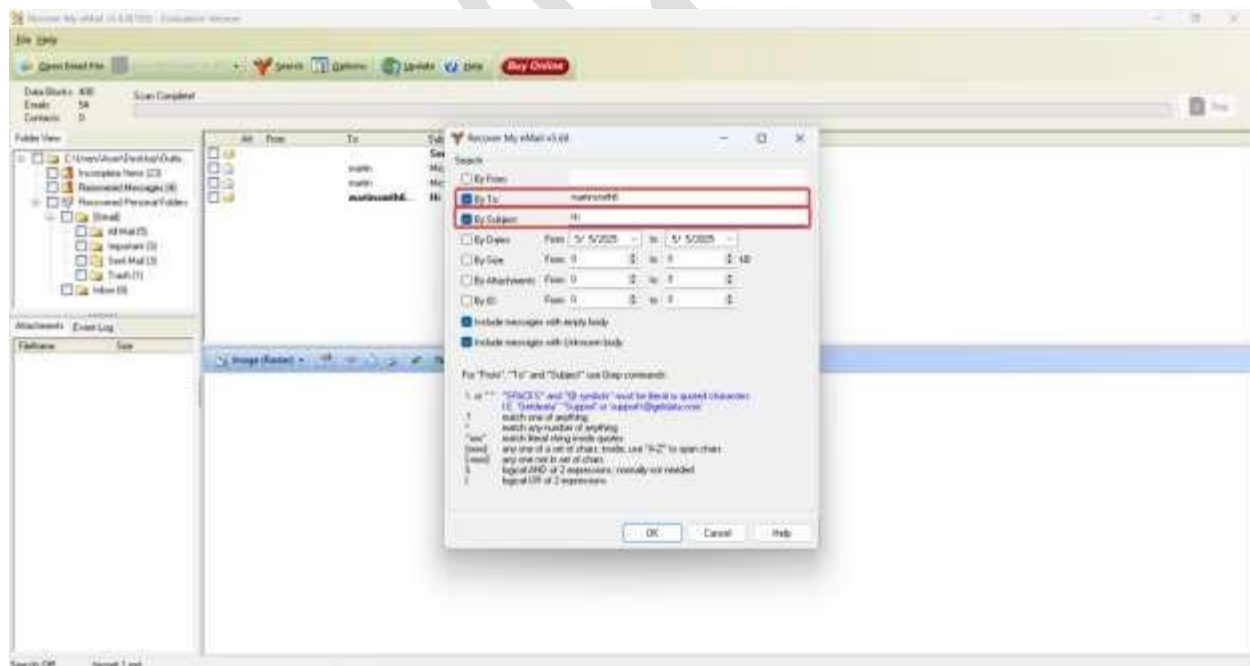


6. To see the emails that have been recovered, click Recovered Messages under the Folder View tab in the left pane of the application window. The application will now display the list of recovered email messages in the right pane of the window. Here, details such as the Subject of the emails, their Received date and time, etc. will be shown.

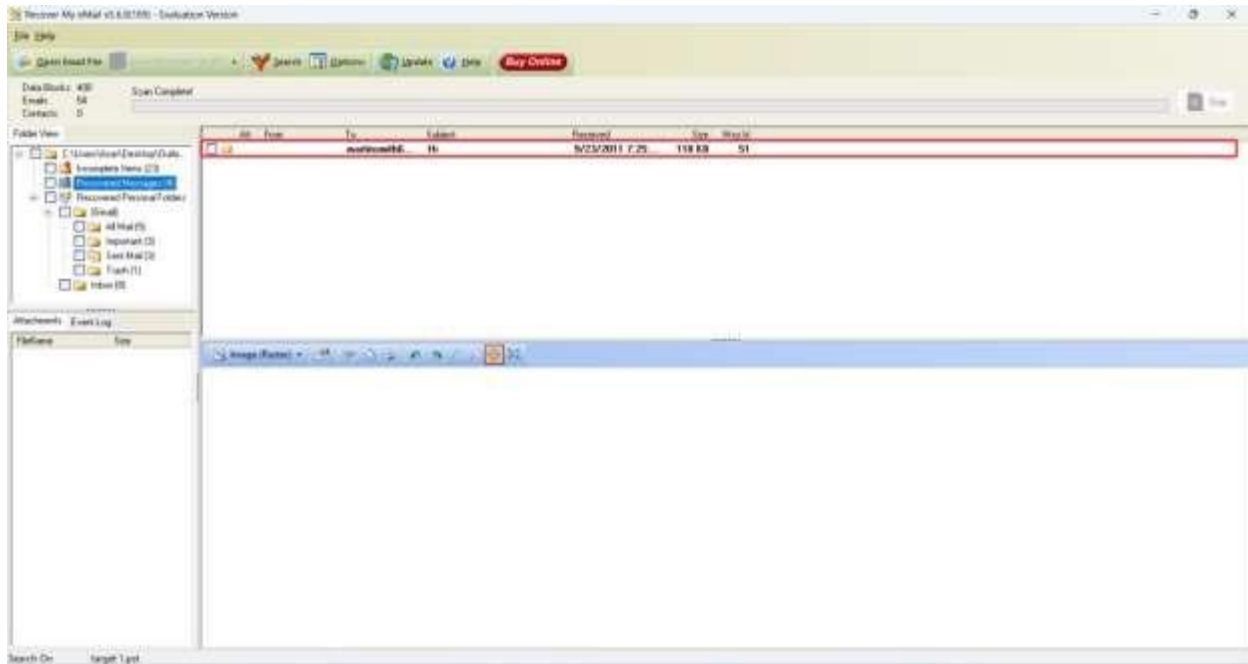


7. In this lab, we are selecting By To: (this field will include a recipient's email ID) and By Subject: (this will include the subject line) options as our filtering criteria.

8. We will provide the input as **martinsmith6** in the **By To:** field and as **Hi** in the **By Subject:** field (we have already seen the terms that match our search filters in the screenshots above in the recipient's (To) email address and the Subject columns, respectively). After specifying the input in these fields, click OK.

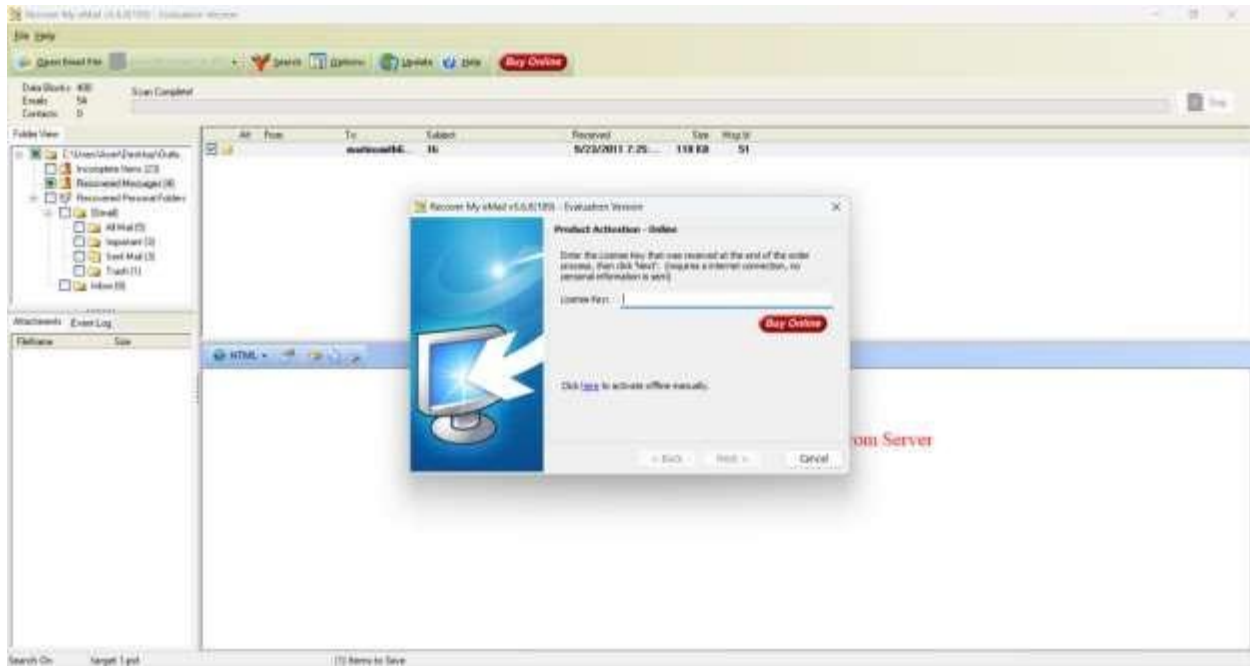


9. Now click the All Mail folder option in the left pane of the window. The application will display the list of emails matching your search criteria in the right pane, as indicated in the screenshot below. In a real scenario, if you see more than one result, then make your search more specific by including other details such as file size, dates, etc., in the search operation.



10. To save the recovered emails, select the checkbox beside the email(s) you want to save and click the Save Message(s) as PST button at the top.

The trial version of the tool does not allow users to save the recovered messages. If you want to save the files, you need to buy the product activation key from the vendor.



11. Close all the windows.

Lab 3: Recover Deleted Email Messages Using Recovery Toolbox for Outlook

Lab Scenario

A brokerage firm has filed a complaint with the district authorities stating that one of its employees has been sharing their trade secrets as well as contract information with their rivals through emails. The firm, though aware of the suspected employee's activities, could not prove her guilty due to the lack of evidence on the emails she was sending out to the rivals. This was because all the emails she had been sending and then receiving in response were being deleted by her in order to evade detection. The court of law has now ordered a probe, which includes scanning of individual devices for emails containing suspicious information. Jason, the forensic investigator, is now assigned with the task of recovering the deleted emails to find out what information has been leaked out of the firm and to prove the suspect's crime and guilt.

Lab Objectives

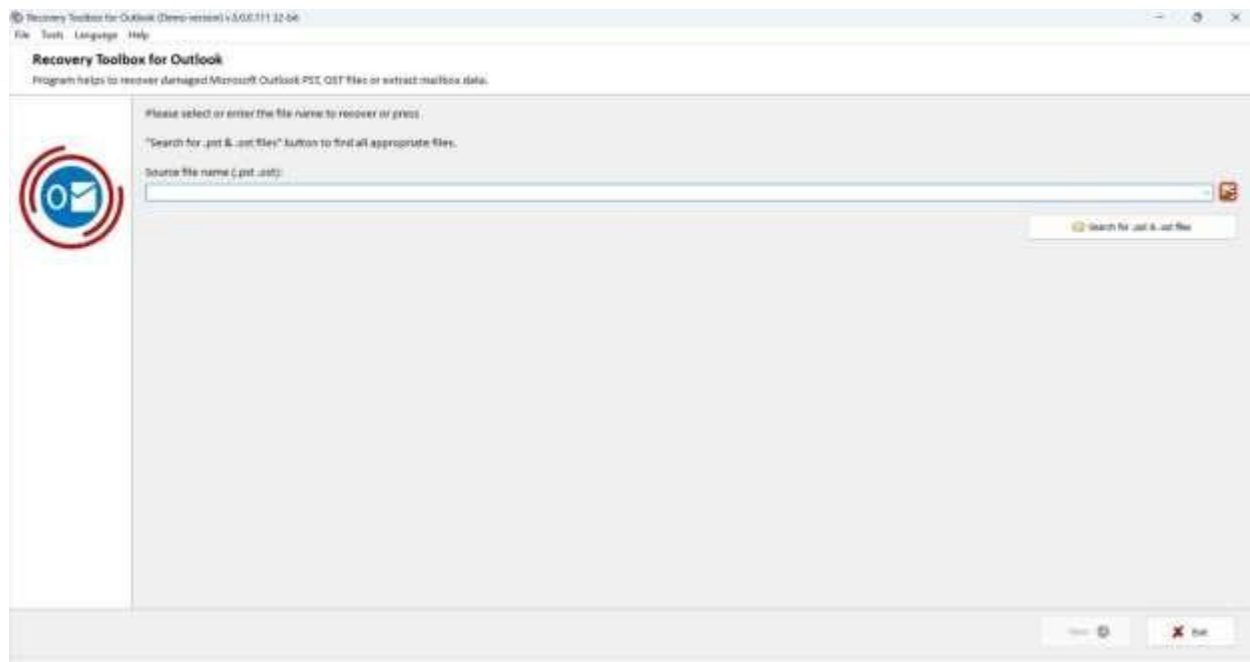
The objective of this lab is to help investigators understand how to recover deleted email messages and attachments using Recovery Toolbox for Outlook.

Overview of the Lab

This lab familiarizes you with the process of searching and retrieval of deleted emails using Recovery Toolbox for Outlook.

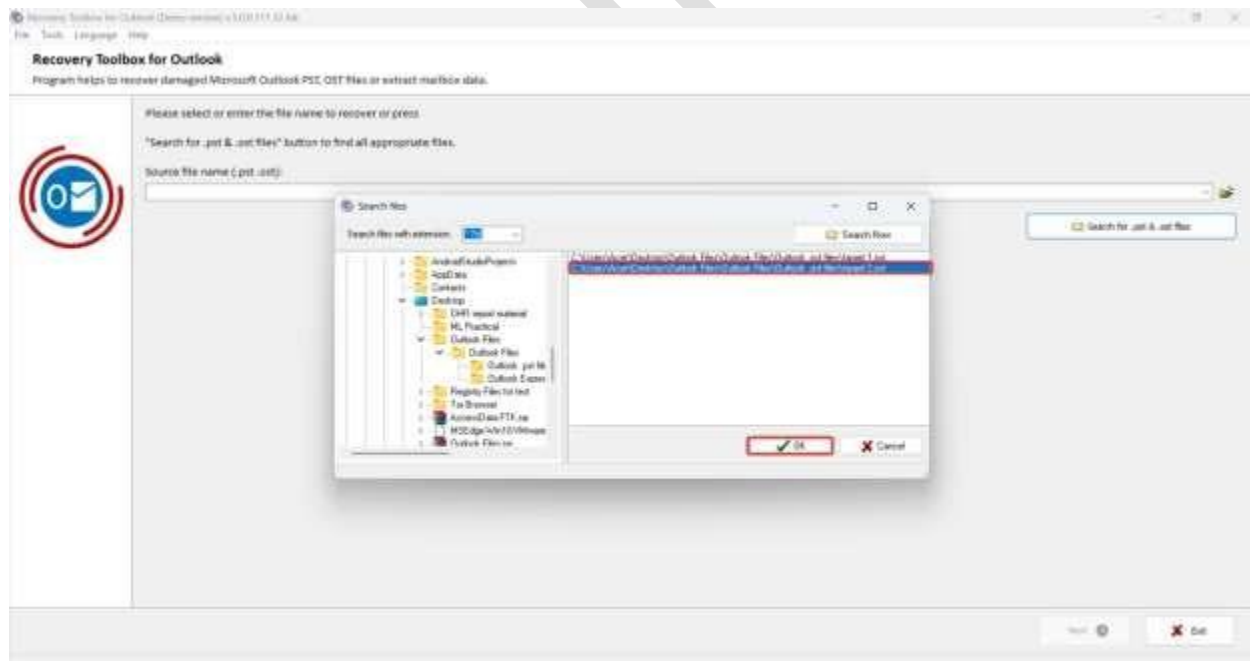
Lab Tasks

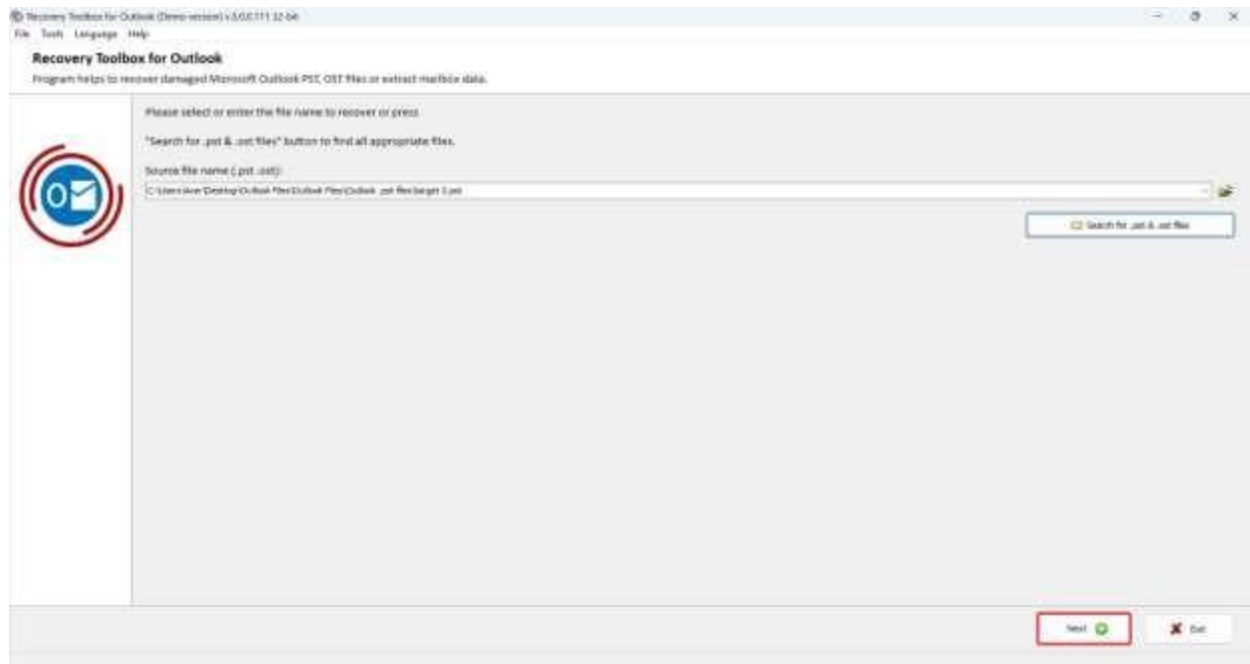
1. Launch Recovery Toolbox for Outlook.
2. The Recovery Toolbox for Outlook window appears, click on the browse icon present below the Source file name (.pst .ost): as shown in the screenshot.



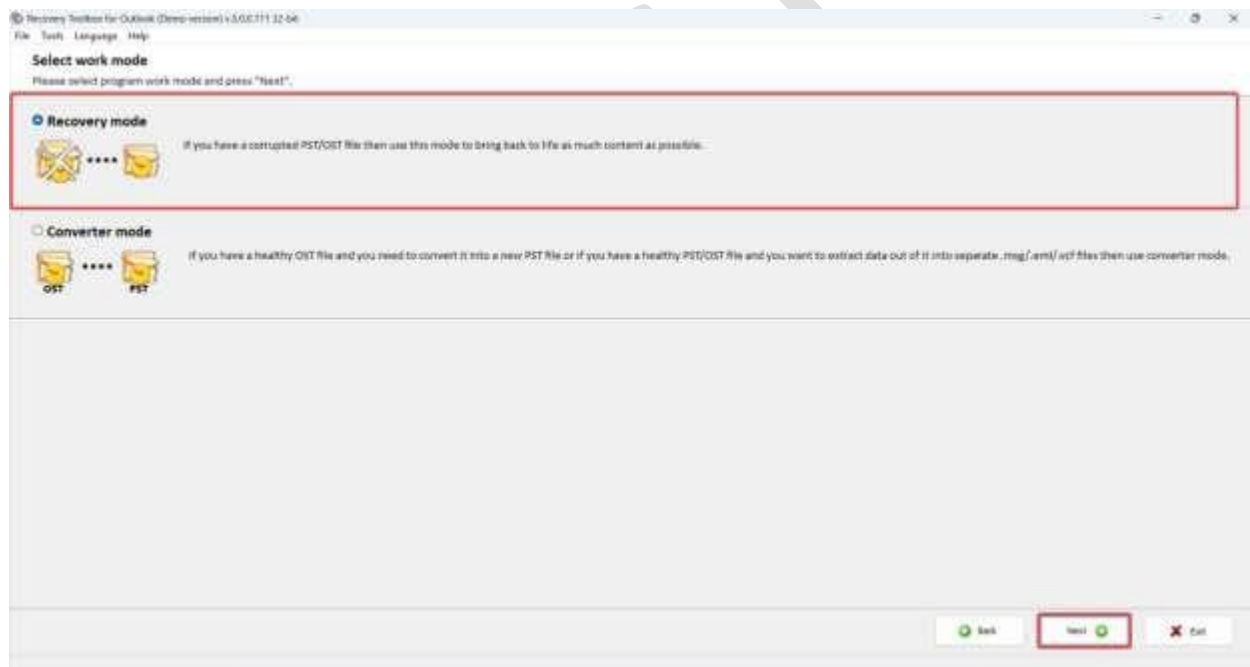
3. In the Select source PST or OST file for recovery window, click on the browse button present under Please select the source PST or OST file for recovery.

4. The Browse window will appear; navigate to C:\User\Acer\Desktop\Outlook Files\Outlook .pst files, select the target 2.pst file, and then click Open

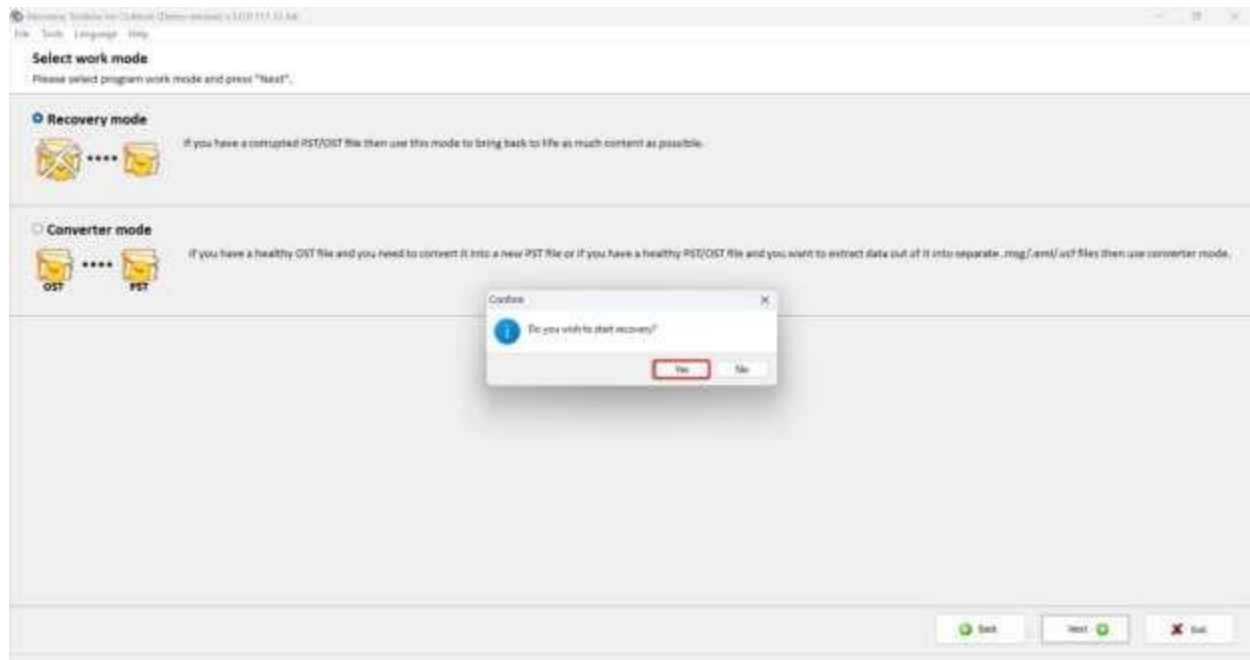




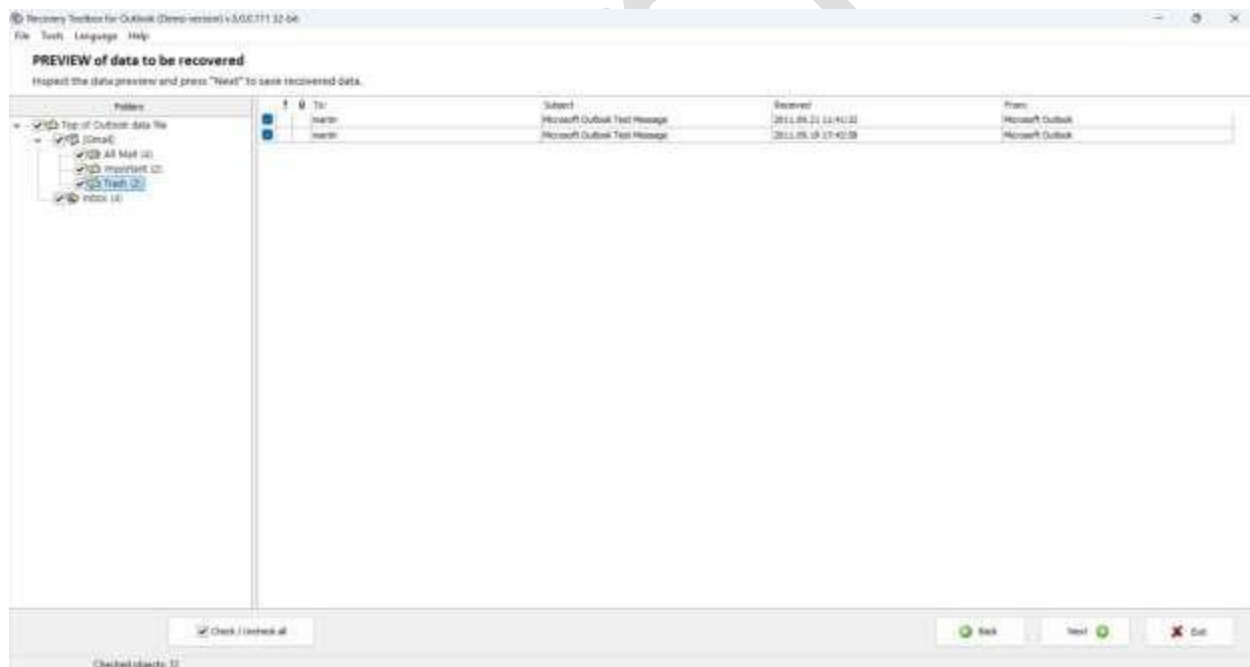
5. Now select Recovery Mode to recover the emails.



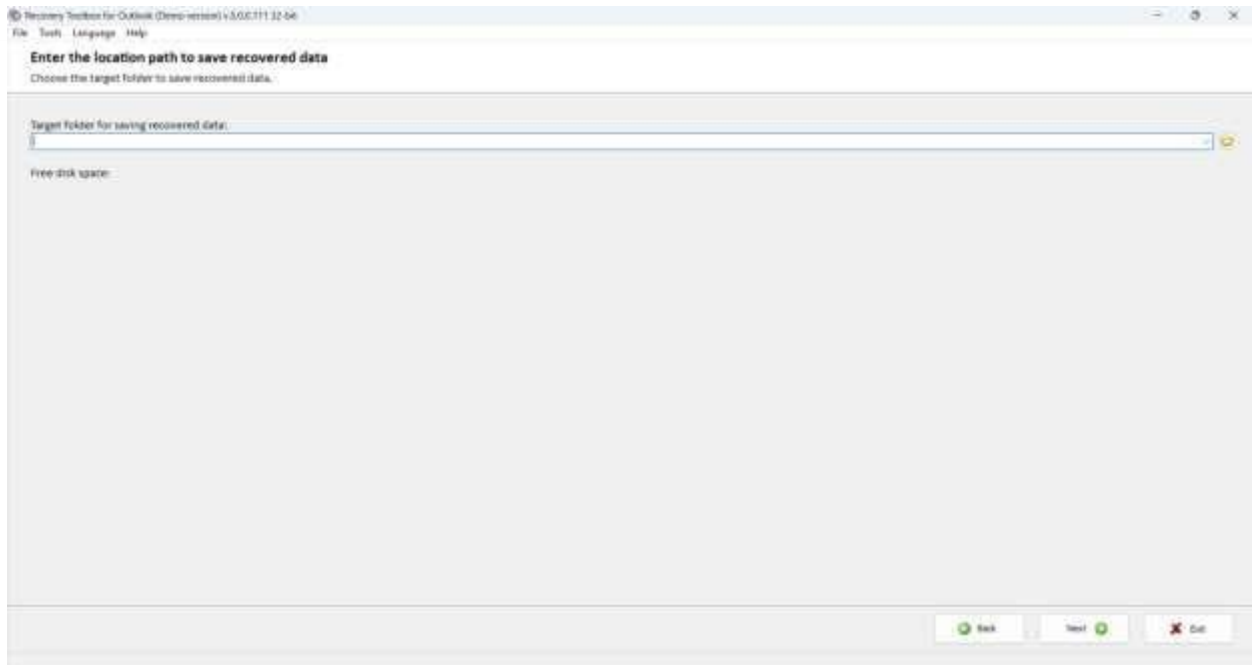
6. When the Confirmation window appears, click OK to continue.



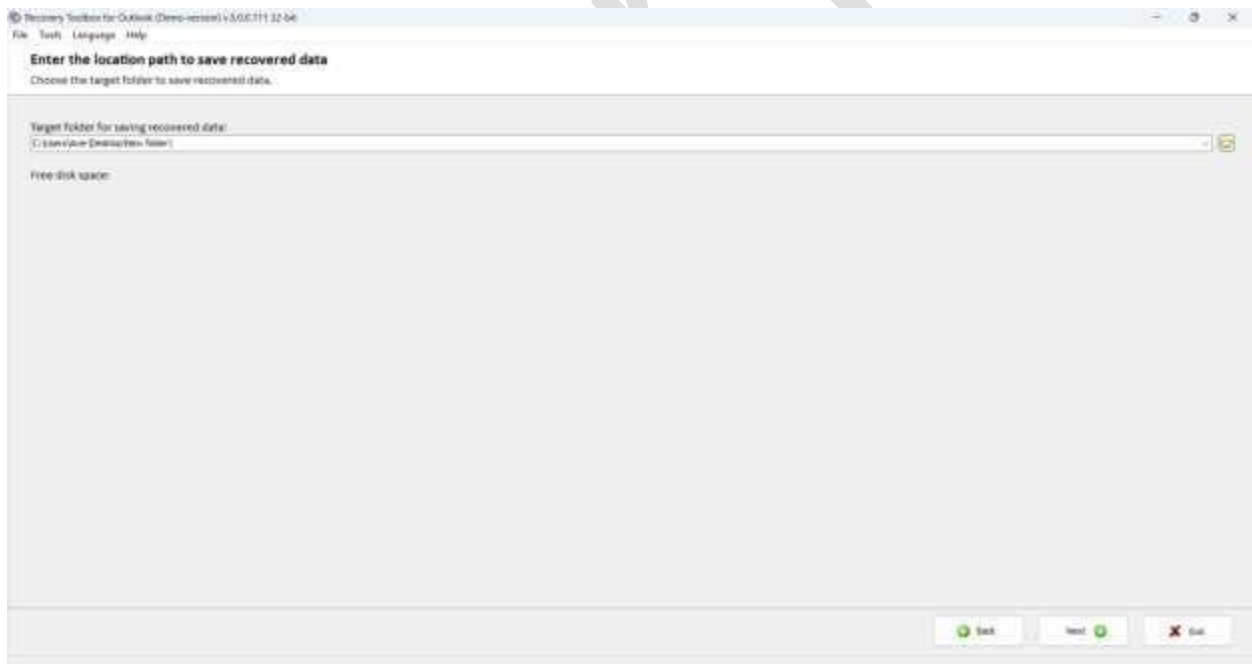
7. The deleted mails will be recovered and previewed in the PREVIEW of data to be recovered. Click on Next.



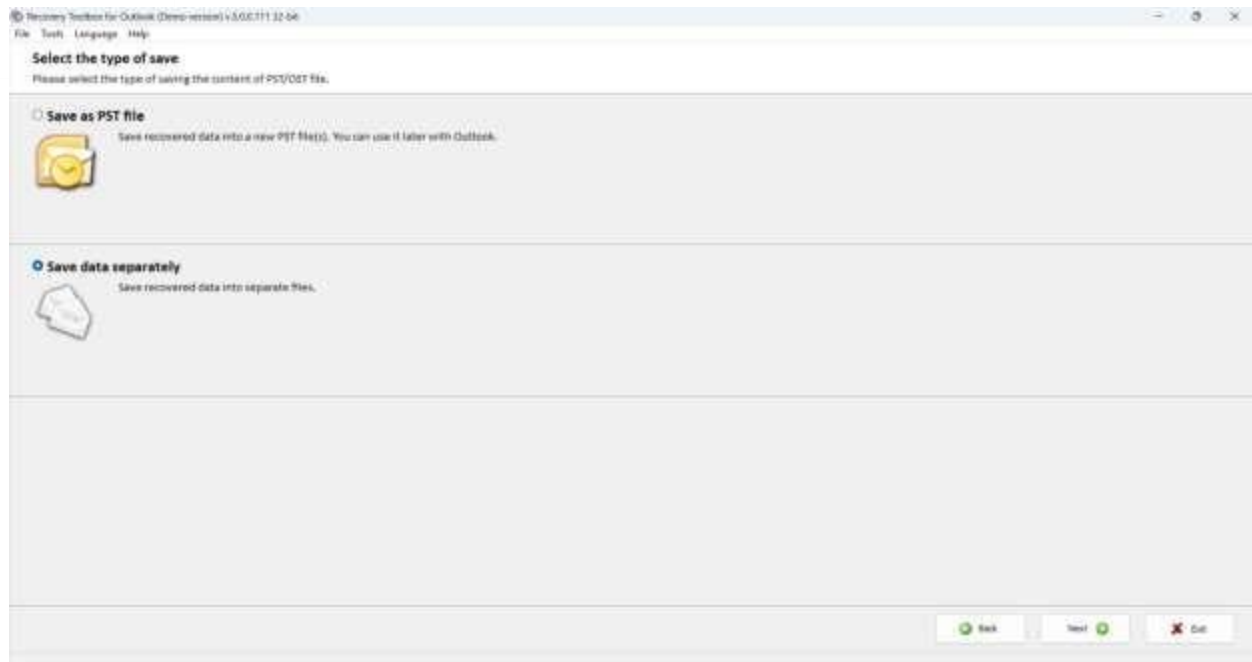
8. Click on the browse button beside the Target folder for saving recovered data:.



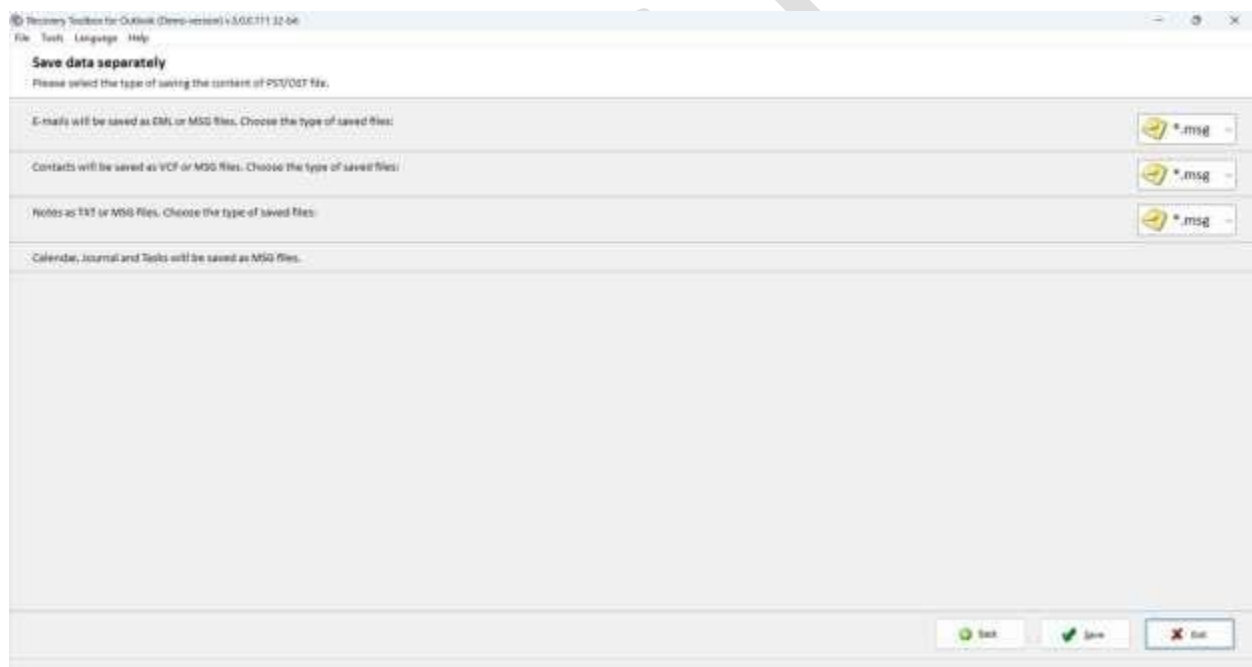
9. In the Browse for Computer window, select the New folder from the Desktop and click on OK.



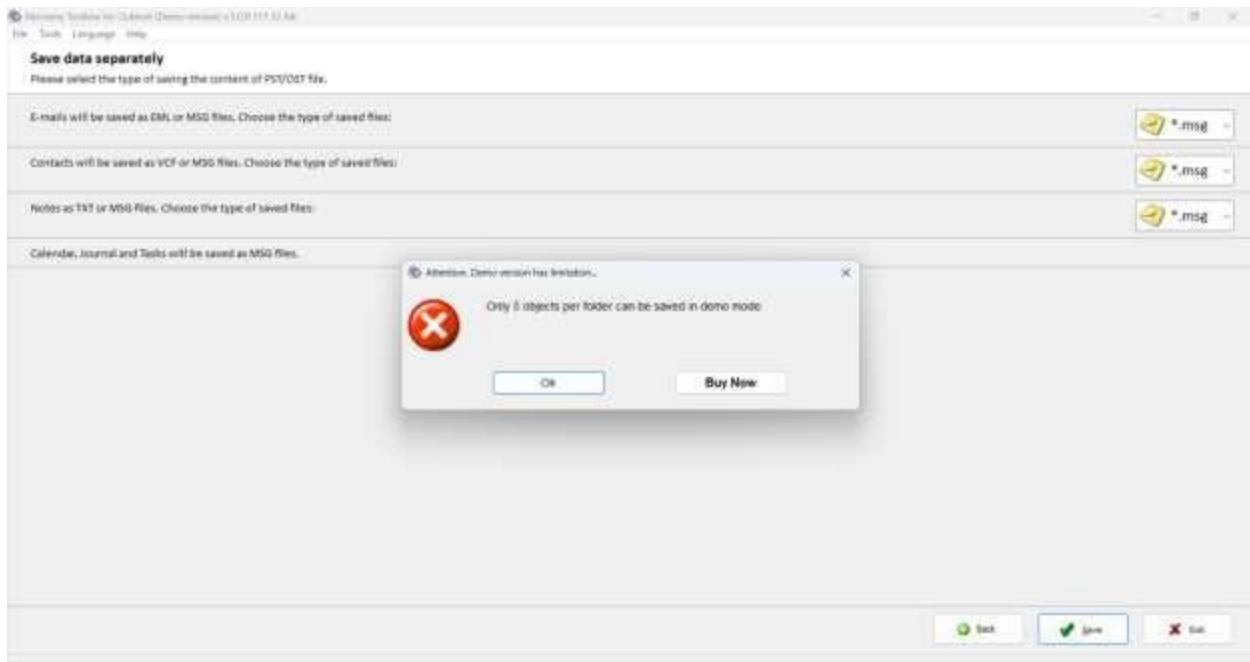
10. In the Select the type of save window, select the Save data separately radio button and click on Next.



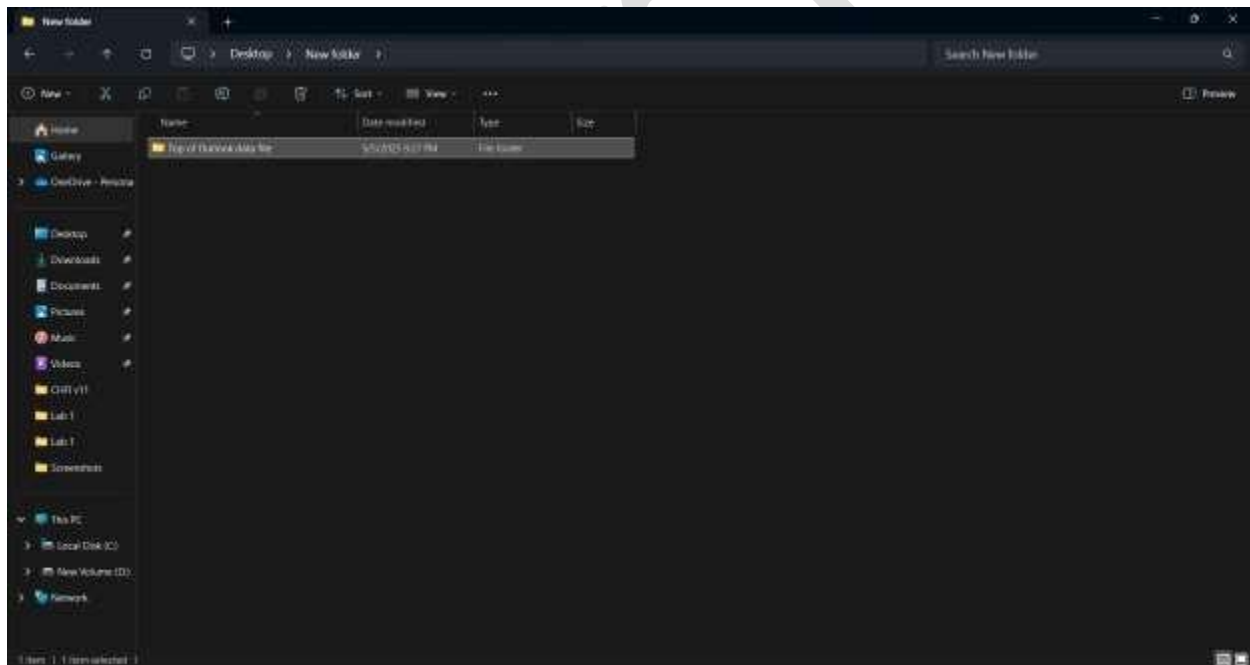
11. In the next window, click on Save to save the recovered email files.



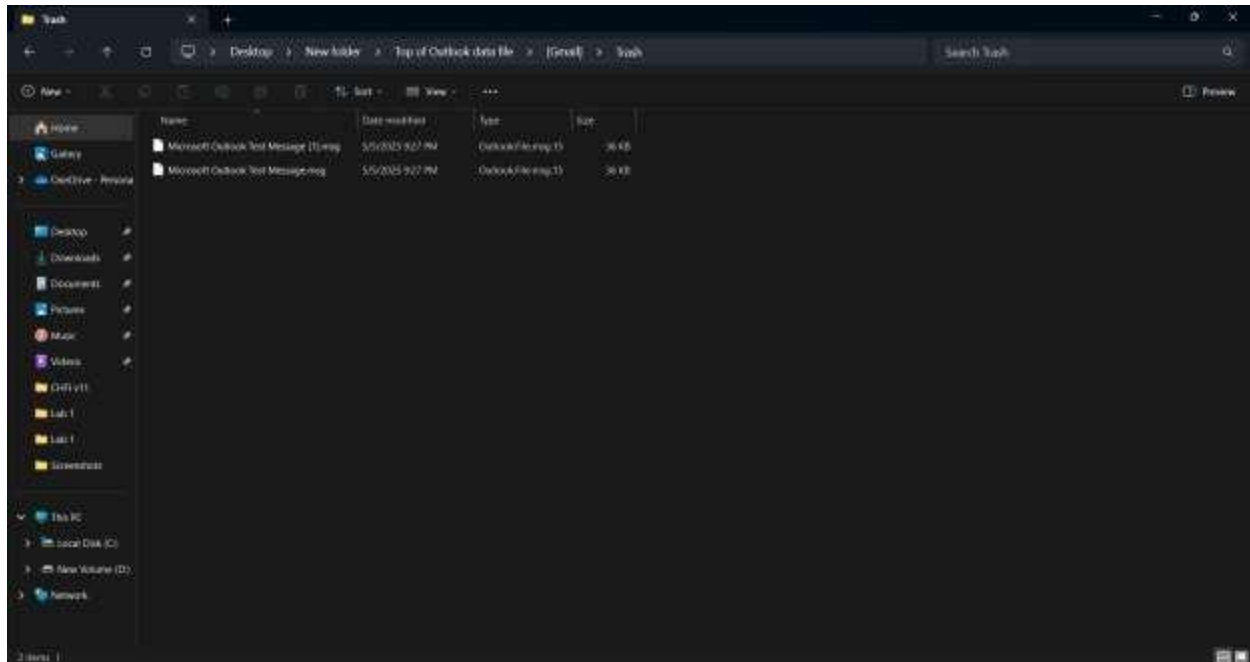
12. In the Attention, Demo version has limitations... window click on OK.



13. The tool will save the email files to the Recovered Email files folder. The File Manager window opens showing the Recovered Email files folder.



14. Open the Recovered mail items folder to view the recovered emails separately.



15. Thus, in this manner, you can recover the deleted emails from .pst files.

16. Close all the windows.

Lab 4: Extract Information from EML Files Using Python

Lab Scenario

A corporate entity has reported suspicious email activity involving potential data exfiltration. The incident response team has acquired a set of EML files associated with the suspicious emails. Thomas, a digital forensic investigator, is tasked with extracting relevant information from these EML files to uncover details about the incident, including sender information, timestamps, and any embedded threats.

Lab Objectives

The objective of this lab is to provide hands-on experience in forensic investigation techniques related to extracting information from EML files using Python.

Overview of the Lab

This lab familiarizes you with the process of extracting information from EML files using Python.

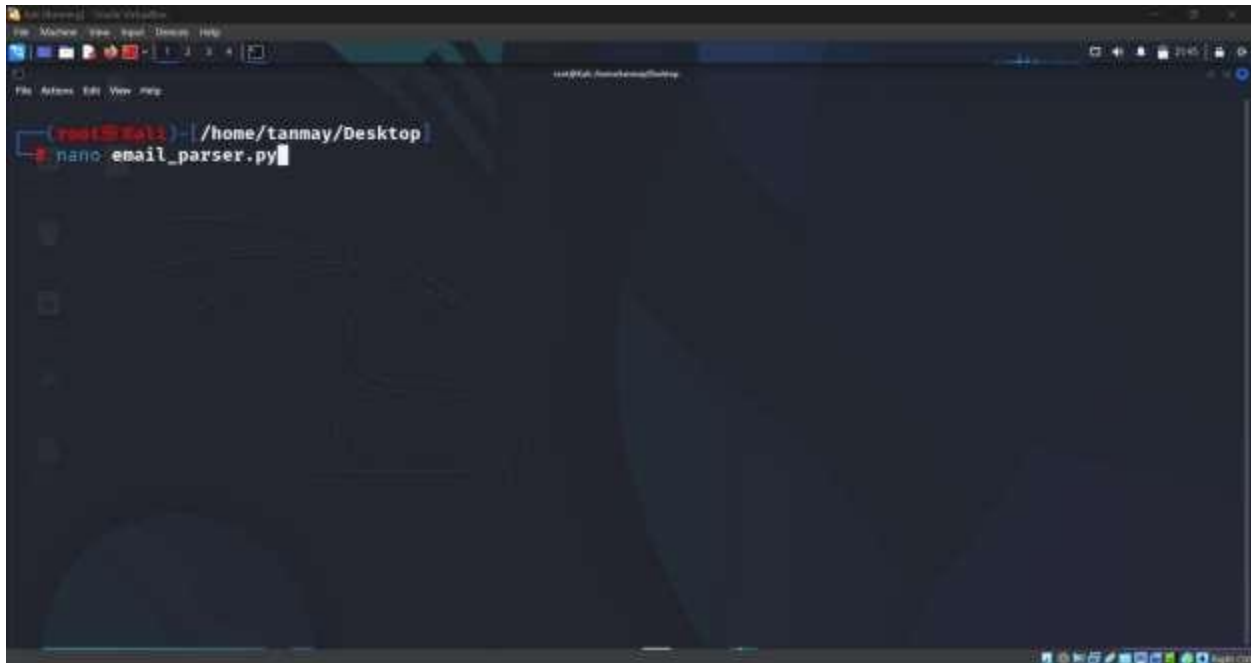
Lab Tasks

1. Launch Kali Linux from Oracle VirtualBox.
2. We can see the Hiii.eml file on the desktop from which we will extract the information as well as the Email_Parser code.

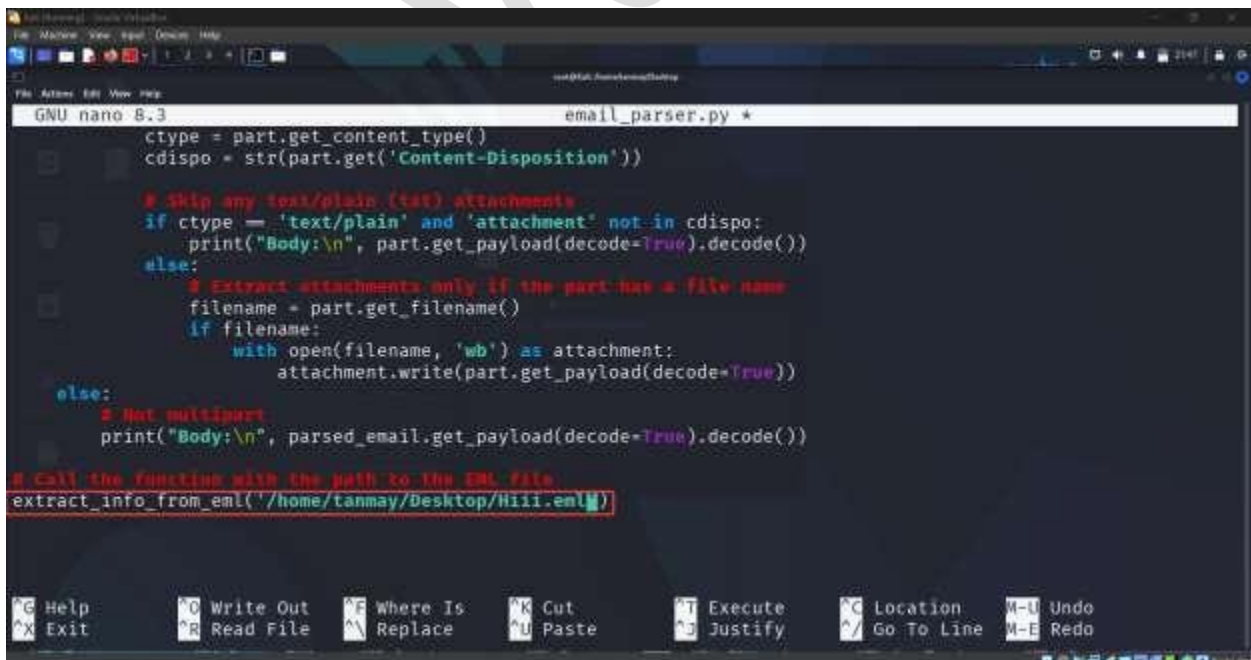


3. Now, click on the Terminal icon from the launcher panel to launch the command-line terminal.

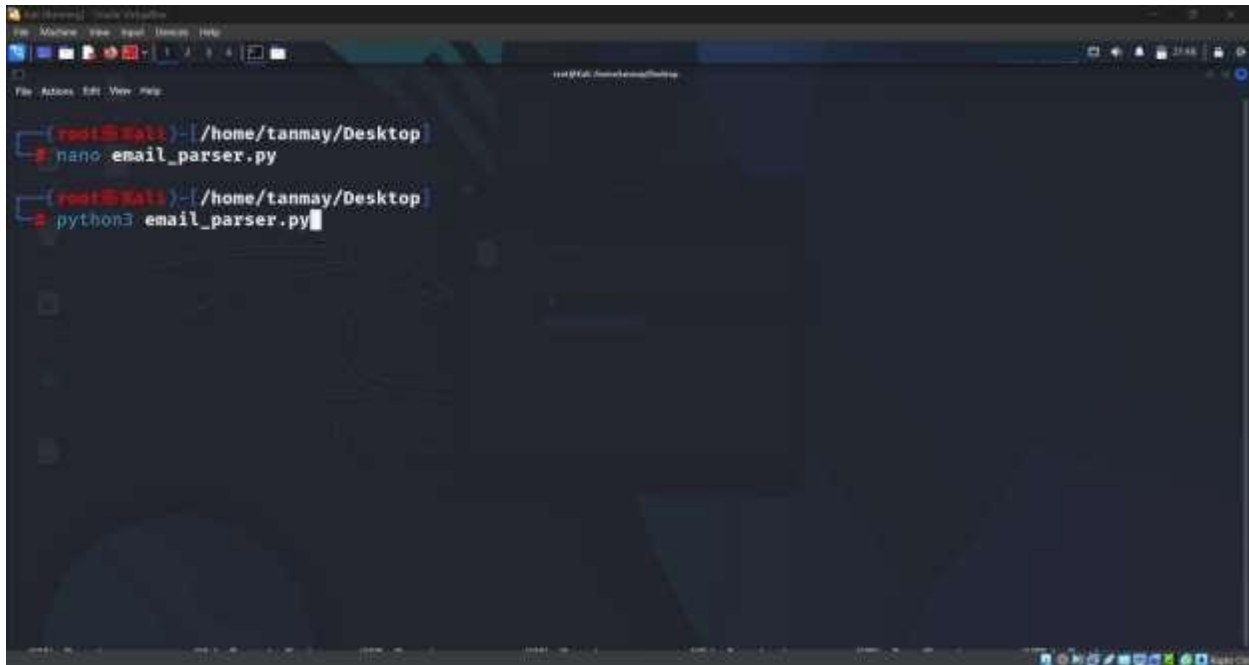
4. Now, we will extract information from EML files using Python script. In the terminal window, type `nano email_parser.py` and press Enter to open a text editor window.



5. To analyze the eml file, we need to replace `path_to_eml_file.eml` in line 32 with the path to `/home/tanmay/Desktop/Hiii.eml` file as shown in the screenshot. Save it and close the Text Editor window.



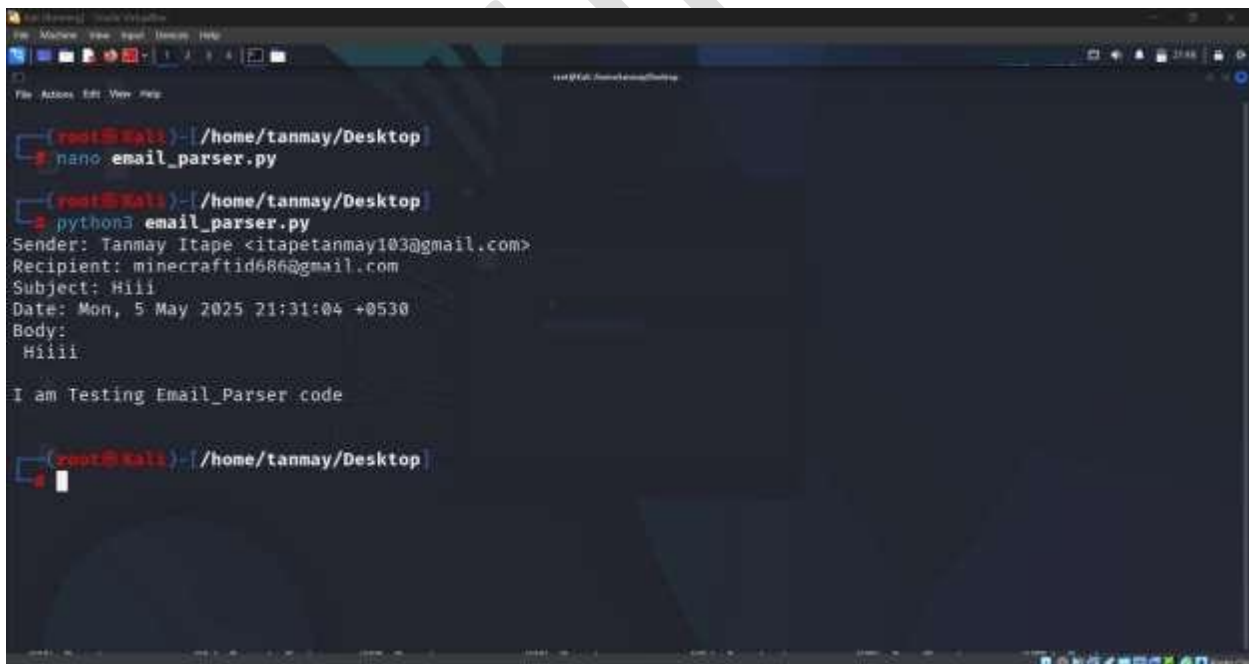
6. Now, switch back to the terminal window, and type `python3 email_parser.py` and press Enter to run the Python script.



```
root@kali: ~/Desktop
# nano email_parser.py

root@kali: ~/Desktop
# python3 email_parser.py
```

7. Upon successful execution of the script you will be able to view the details and contents of the email, as shown in the screenshot.



```
root@kali: ~/Desktop
# nano email_parser.py

root@kali: ~/Desktop
# python3 email_parser.py
Sender: Tanmay Itape <itapetanmay103@gmail.com>
Recipient: minecraftid686@gmail.com
Subject: Hiii
Date: Mon, 5 May 2025 21:31:04 +0530
Body:
Hiiii

I am Testing Email_Parser code

root@kali: ~/Desktop
#
```

8. Thus, in this manner, you can extract information from an eml file using Python.

9. Close all the windows.

Mayur Patil